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Lab 8

1.

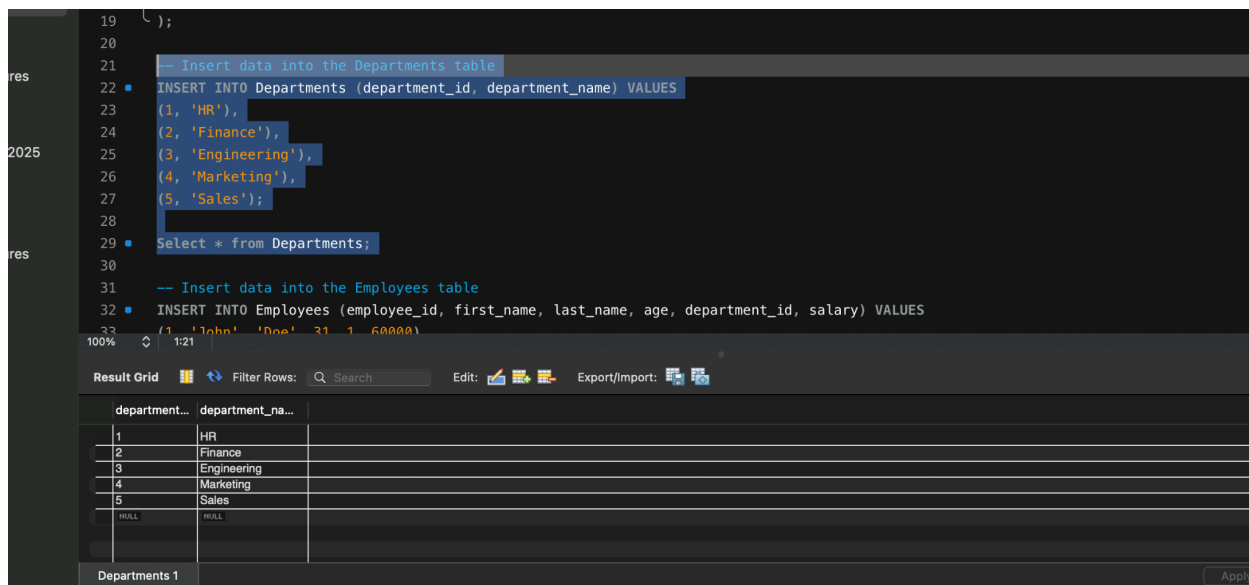
-- Create the Departments table

```
CREATE TABLE Departments (  
    department_id INT PRIMARY KEY,  
    department_name VARCHAR(50) NOT NULL);
```

-- Insert data into the Departments table

```
INSERT INTO Departments (department_id, department_name) VALUES  
(1, 'HR'),  
(2, 'Finance'),  
(3, 'Engineering'),  
(4, 'Marketing'),  
(5, 'Sales');
```

Select * from Departments;



```
19 );  
20  
21 -- Insert data into the Departments table  
22 INSERT INTO Departments (department_id, department_name) VALUES  
23 (1, 'HR'),  
24 (2, 'Finance'),  
25 (3, 'Engineering'),  
26 (4, 'Marketing'),  
27 (5, 'Sales');  
28  
29 Select * from Departments;  
30  
31 -- Insert data into the Employees table  
32 INSERT INTO Employees (employee_id, first_name, last_name, age, department_id, salary) VALUES  
33 (1, 'John', 'Doe', 31, 1, 60000);
```

Result Grid

department...	department_na...
1	HR
2	Finance
3	Engineering
4	Marketing
5	Sales
NULL	NULL

Departments 1

-- Create the Employees table

```
CREATE TABLE Employees (  
    employee_id INT PRIMARY KEY,  
    first_name VARCHAR(50) NOT NULL,
```

```

last_name VARCHAR(50) NOT NULL,
age INT CHECK (age > 0),
department_id INT,
salary DECIMAL(10,2) CHECK (salary > 0),
FOREIGN KEY (department_id) REFERENCES Departments(department_id));

```

-- Insert data into the Employees table

```

INSERT INTO Employees (employee_id, first_name, last_name, age, department_id, salary) VALUES
(1, 'John', 'Doe', 31, 1, 60000),
(2, 'Robert', 'Luna', 22, 2, 55000),
(3, 'David', 'Robinson', 29, 3, 80000),
(4, 'Betty', 'Smith', 35, 3, 90000),
(5, 'Alice', 'Johnson', 28, 4, 70000),
(6, 'Eve', 'Doe', 32, 5, 65000),
(7, 'Charlie', 'Brown', 27, 3, 75000);

```

```
select * from Employees;
```

The screenshot shows a SQL IDE interface. The top pane contains the following SQL code:

```

29 Select * from Departments;
30
31 -- Insert data into the Employees table
32 INSERT INTO Employees (employee_id, first_name, last_name, age, department_id, salary) VALUES
33 (1, 'John', 'Doe', 31, 1, 60000),
34 (2, 'Robert', 'Luna', 22, 2, 55000),
35 (3, 'David', 'Robinson', 29, 3, 80000),
36 (4, 'Betty', 'Smith', 35, 3, 90000),
37 (5, 'Alice', 'Johnson', 28, 4, 70000),
38 (6, 'Eve', 'Doe', 32, 5, 65000),
39 (7, 'Charlie', 'Brown', 27, 3, 75000);
40
41 select * from Employees;
42
43
44
45

```

The bottom pane shows the 'Result Grid' with the following data:

employee_id	first_name	last_name	age	department...	salary
1	John	Doe	31	1	60000.00
2	Robert	Luna	22	2	55000.00
3	David	Robinson	29	3	80000.00
4	Betty	Smith	35	3	90000.00
5	Alice	Johnson	28	4	70000.00
6	Eve	Doe	32	5	65000.00
7	Charlie	Brown	27	3	75000.00
	NULL	NULL	NULL	NULL	NULL

Below the grid, there is a label 'Employees 2' and an 'Apply' button.

3.

Exercise 1:

```

select e.employee_id, e.first_name, e.last_name, e.age, e.salary, d.department_name
from Employees e
join Departments d
ON e.department_id = d.department_id

```

```
where d.department_name = 'Engineering';
```

```
36 (4, 'Betty', 'Smith', 35, 3, 90000),
37 (5, 'Alice', 'Johnson', 28, 4, 70000),
38 (6, 'Eve', 'Doe', 32, 5, 65000),
39 (7, 'Charlie', 'Brown', 27, 3, 75000);
40
41 • select * from Employees;
42
43 • select e.employee_id, e.first_name, e.last_name, e.age, e.salary, d.department_name
44 from Employees e
45 join Departments d
46 ON e.department_id = d.department_id
47 where d.department_name = 'Engineering';
48
49
50
51
```

100% 41:47

Result Grid Filter Rows: Search Export:

employee_id	first_name	last_name	age	salary	department_na...
3	David	Robinson	29	80000.00	Engineering
4	Betty	Smith	35	90000.00	Engineering
7	Charlie	Brown	27	75000.00	Engineering

Result 3

Ex.2

```
select e.employee_id, e.first_name, e.last_name, e.age, e.salary, d.department_name
from employees e
join departments d
on e.department_id = d.department_id
where e.salary > (
select avg(e2.salary)
from employees e2
where e2.department_id = e.department_id);
```

```
47 where d.department_name = 'Engineering';
48
49
50
51 • select e.employee_id, e.first_name, e.last_name, e.age, e.salary, d.department_name
52 from employees e
53 join departments d
54 on e.department_id = d.department_id
55 where e.salary > (
56   select avg(e2.salary)
57   from employees e2
58   where e2.department_id = e.department_id);
59
60
61
62
```

100% 43:58

Result Grid Filter Rows: Search Export:

employee_id	first_name	last_name	age	salary	department_na...
4	Betty	Smith	35	90000.00	Engineering

Result 4

Ex.3

```
select d.department_name, sum(e.salary) as total_salary
from employees e
join departments d
on e.department_id = d.department_id
group by d.department_name;
```

```
52 from employees e
53 join departments d
54 on e.department_id = d.department_id
55 where e.salary > (
56   select avg(e2.salary)
57   from employees e2
58   where e2.department_id = e.department_id);
59
60
61 • select d.department_name, sum(e.salary) as total_salary
62 from employees e
63 join departments d
64 on e.department_id = d.department_id
65 group by d.department_name;
66
67
```

100% 1:66

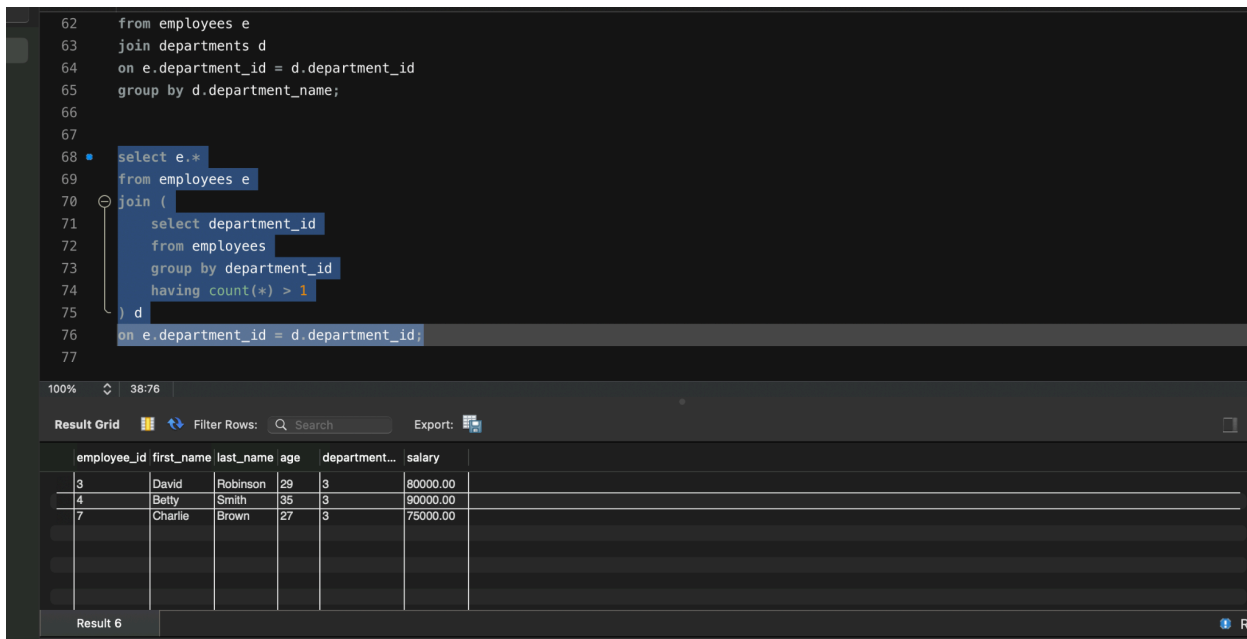
Result Grid Filter Rows: Search Export:

department_na...	total_salary
HR	80000.00
Finance	55000.00
Engineering	245000.00
Marketing	70000.00
Sales	65000.00

Result 5

Ex.4

```
select e.*
from employees e
join (
  select department_id
  from employees
  group by department_id
  having count(*) > 1
) d
on e.department_id = d.department_id;
```



The screenshot shows a SQL IDE with a query editor and a results grid. The query in the editor is as follows:

```
62 from employees e
63 join departments d
64 on e.department_id = d.department_id
65 group by d.department_name;
66
67
68 select e.*
69 from employees e
70 join (
71   select department_id
72   from employees
73   group by department_id
74   having count(*) > 1
75 ) d
76 on e.department_id = d.department_id;
77
```

The results grid displays the following data:

employee_id	first_name	last_name	age	department...	salary
3	David	Robinson	29	3	80000.00
4	Betty	Smith	35	3	90000.00
7	Charlie	Brown	27	3	75000.00

The results grid also shows a search bar, a filter row, and an export button. The status bar at the bottom indicates "Result 6".

Ex.5

```
select d.department_name
from employees e
join departments d
on e.department_id = d.department_id
where e.salary = (select max(salary) from employees);
```

```
75 ) d
76 on e.department_id = d.department_id;
77
78
79
80 select d.department_name
81 from employees e
82 join departments d
83 on e.department_id = d.department_id
84 where e.salary = (select max(salary) from employees);
85
86
87
88
89
90
```

100% 54:84

Result Grid Filter Rows: Search Export:

department_na...
Engineering

Result 7

Ex.6

```
select d.department_name, avg(e.age) as average_age
from employees e
join departments d
on e.department_id = d.department_id
group by d.department_name;
```

```
72 from employees
73 group by department_id
74 having count(*) > 1
75 ) d
76 on e.department_id = d.department_id;
77
78
79
80 select d.department_name
81 from employees e
82 join departments d
83 on e.department_id = d.department_id
84 where e.salary = (select max(salary) from employees);
85
86
87 select d.department_name, avg(e.age) as average_age
88 from employees e
89 group by d.department_name;
```

100% 28:91

Result Grid Filter Rows: Search Export:

department_na...	average_age
HR	31.0000
Finance	22.0000
Engineering	30.3333
Marketing	28.0000
Sales	32.0000

Result 8

Ex.7

```
select e.*  
from employees e  
join departments d  
on e.department_id = d.department_id  
where d.department_name in ('HR', 'Finance');
```

```
84  where e.salary = (select max(salary) from employees);  
85  
86  
87  • select d.department_name, avg(e.age) as average_age  
88      from employees e  
89      join departments d  
90      on e.department_id = d.department_id  
91      group by d.department_name;  
92  
93  • select e.*  
94      from employees e  
95      join departments d  
96      on e.department_id = d.department_id  
97      where d.department_name in ('HR', 'Finance');  
98  
99
```

100% 46:97

Result Grid Filter Rows: Search Export:

employee_id	first_name	last_name	age	department...	salary
1	John	Doe	31	1	60000.00
2	Robert	Luna	22	2	55000.00

Result 9